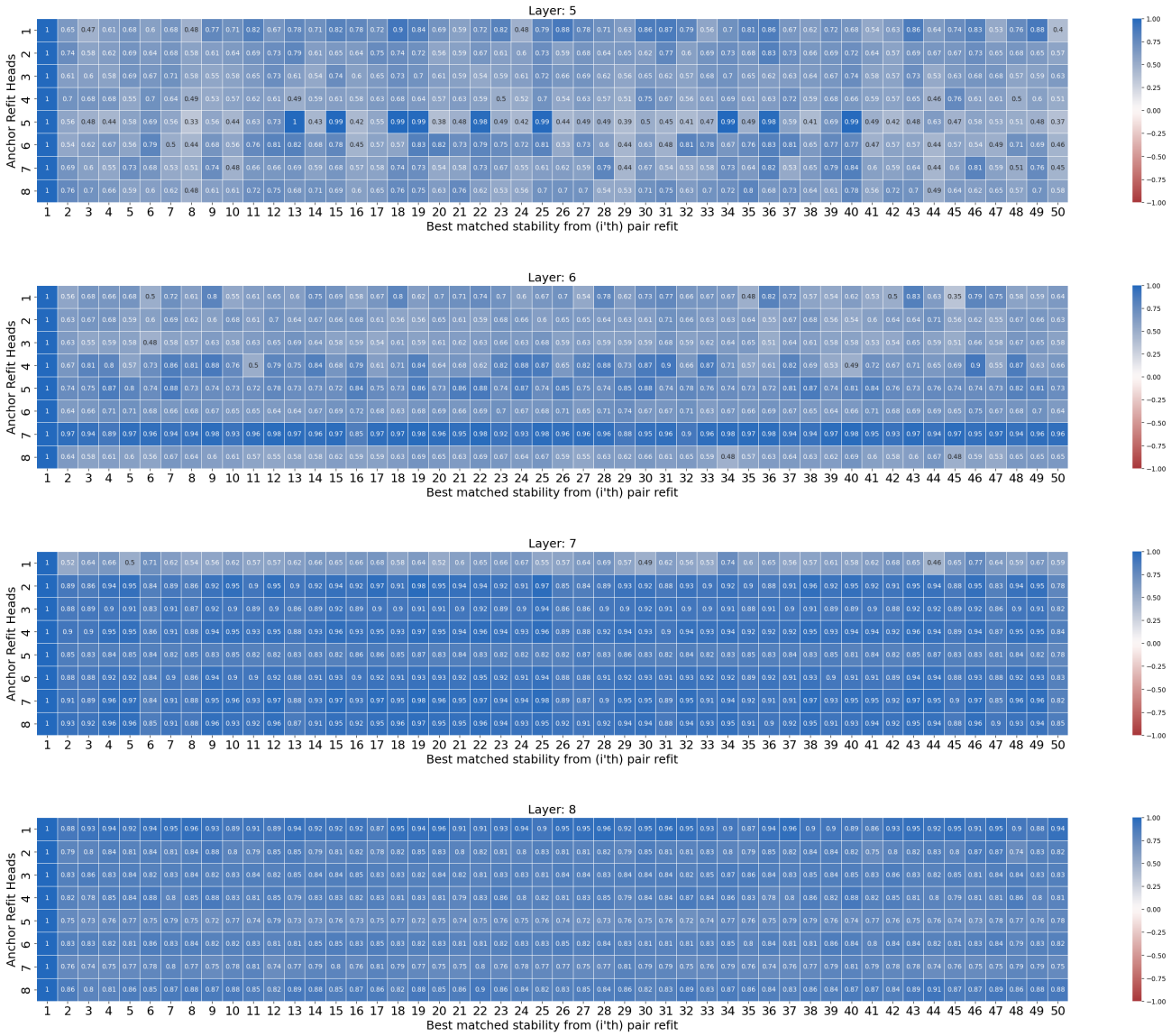


Figure 12. Residual stream is relatively more stable than its corresponding attention heads.

A.6. Within-layer uniqueness of attention heads

Methodology to calculate commonness (non-uniqueness) of an attention head h_i (See § 4.3):

Setup and notation

- Let $\mathcal{P}$ be a fixed set of 100 prompts.
- h_i is a head belonging to layer l in a refit m.
- h_j can be any head belonging to layer l in the same refit m.
- All possible h_j heads are considered as *peer heads* to be compared with.

A.6.1. Step 1: Prompt-wise head similarity

For a head h_i of layer l in anchor refit m and one of its peer heads h_j, we define the prompt-wise similarity between h_i and h_j as the *cosine similarity* between their vectorized (flattened) attention score matrices for prompt $p \in \mathcal{P}$, that is

$((A^m_{h_i}(p)), (A^m_{h_j}(p)))$, from heads h_i and h_j respectively:

$$C^{(m)}_{(h_i,h_j)}(p) = CosineSim\left(\text{vec}\big(A^m_{h_i}(p)\big), \text{vec}\big(A^m_{h_j}(p)\big)\right) \tag{8}$$

A.6.2. Step 2: Average similarity score across prompts

We aggregate over prompts to obtain an average similarity scores between the two heads (h_i and h_j):

$$\bar{C}^{(m)}_{(h_i,h_j)} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} C^{(m)}_{(h_i,h_j)}(p) \tag{9}$$

A.6.3. Step 3: Commonness of a head

By repeating Step 2 (Eq. 9), We compare head h_i against each peer heads $h_j \in \{1, \ldots, H^m\}$. Averaging these head-to-head comparisons yields per-head redundancy score, quantifying how common (or unique) h_i is within its layer:

$$Commonness^{(m)}_{(h_i)} = \frac{1}{Number\ of\ heads} \sum_{h_j \in \{1,\ldots,H^m\}} \bar{C}^{(m)}_{(h_i,h_j)} \tag{10}$$

A.6.4. Step 4: Average redundancy of a layer

The average of calculated commonness from Step 3 (Eq. 10) for all the heads $h_i \in \{1, \ldots, H^m\}$ provides a sense of average redundancy within the layer:

$$Commonness^{(m)}_{l} = \frac{1}{Number\ of\ heads} \sum_{h_j \in \{1,\ldots,H^m\}} \bar{Commonness}^{(m)}_{(h_i)} \tag{11}$$

We can now plot average redundancy for all layers of the architecture at hand, which shows that middle layer attention heads are more unique, relative to those in early and later layers (See Fig. 4 and § B.3)

A.7. Stability of residual stream

We followed the CKA implementation provided in (Kornblith et al., 2019) to compare layer counterparts of different refits using their set of activations over the prompt set $\mathcal{P}$. To compute centered Gram matrices we used an RBF kernel with a threshold of 1.0.

A.8. Layer-wise correlation between the stability and "post-ablation change in perplexity"

To probe implications of head instability, we conduct a head ablation study on the anchor refit. For each layer l, we:

1. For all the heads $h_i \in l$, we calculate the head's stability (S_{h_i}).
2. For all the heads $h_i \in l$, we measure the post-ablation change in perplexity.

We define the "post-ablation change in perplexity" as change in perplexity after ablating the head h_i, that is, zero'ing the head h_i output during the forward pass. The perplexity was calculated based on a 100-prompts set § A.1:

$$\Delta\text{PPL}_{\text{h}_\text{i}} = \text{PPL}_{\text{ablated}}(h_i) - \text{PPL}_{\text{baseline}} \tag{12}$$

For each layer, we then compute the correlation between two sets containing the:

1. $\{S_{h_i}\}^H_{h_i=1}$: Stability for all the heads $h_i \in$ layer l
2. $\{\Delta\text{PPL}_{\text{h}_\text{i}}\}^H_{h_i=1}$: "post-ablation change in perplexity" for all the heads $h_i \in$ layer l

A.9. Geometric observation of the attention head activations using meta-SNE

Method for calculating meta-SNE embeddings:

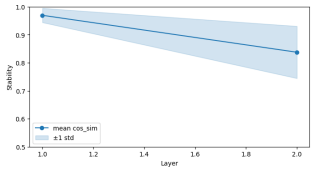
Following the approach laid out in (Olah, 2015), for each head h, we represent its behavior by using its output activations over a fixed prompt set $\mathcal{P}$. We compute a pairwise distance matrix D whose $D_{i,j}$ entry reflect the dissimilarity between pairs of points within the representation space of head h. The points are attention head h output activation for different prompts in $\mathcal{P}$. We use Euclidean distance to calculate the distance matrix D. Because meta-SNE consumes distances rather than raw coordinates, it is insensitive to many isometric transformations (common in representation spaces), preserving both local and global neighborhood structure. The resulting distance matrix is then embedded with t-SNE to produce a 2-D embeddings.

Related figures are provided in § 4.9.

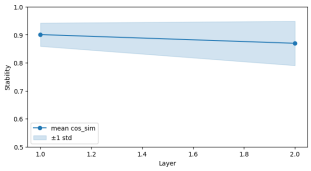
B. Results

B.1. layer-wise stability

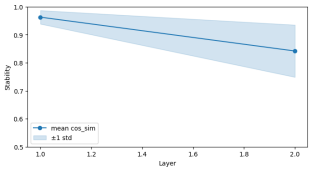
Additional architecture-wise plots for § 4.1:



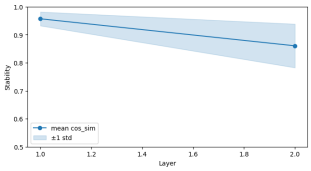
(a) Layers: 2, Heads: 8, Attn-only: False, Adam



(b) Layers: 2, Heads: 16, Attn-only: False, Adam



(c) Layers: 2, Heads: 8, Attn-only: False, AdamW



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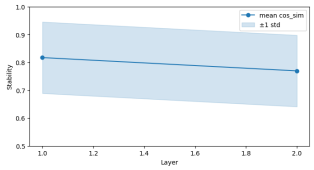
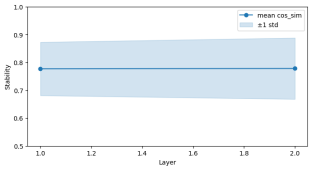


(a) Layers: 2, Heads: 8, Attn-only: True, Adam



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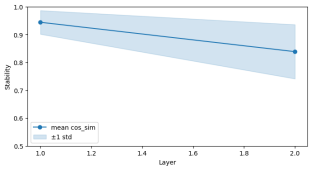

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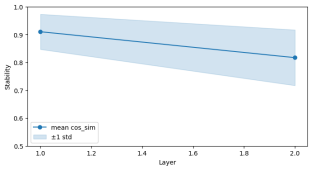

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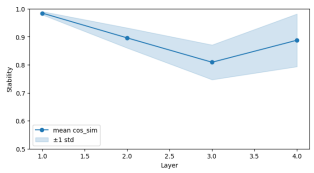

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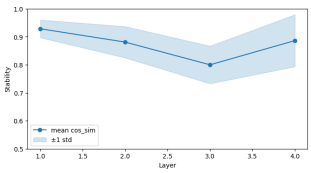

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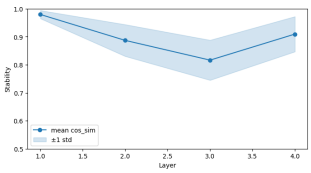

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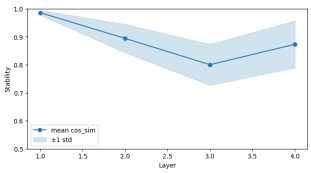
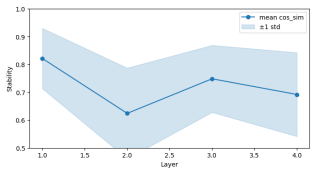
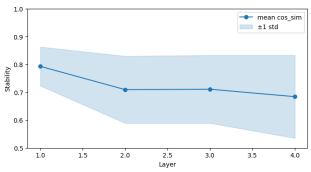


(d) Layers: 4, Heads: 16, Attn-only: False, AdamW



(a) Layers: 4, Heads: 8, Attn-only: True, Adam



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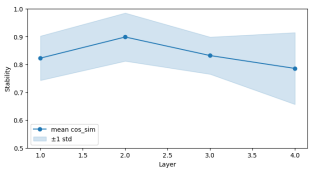

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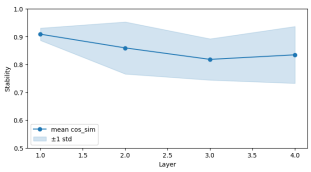

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